SCR2043 OPERATING SYSTEMS

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Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
`onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -e`
Output

```
  PID TTY          TIME CMD
    1 ?        00:00:02 systemd
    2 ?        00:00:00 kthreadd
    3 ?        00:00:00 kworker/0:0
    4 ?        00:00:00 kworker/0:0H
    6 ?        00:00:00 mm_percpu_wq
    7 ?        00:00:00 ksoftirqd/0
    8 ?        00:00:00 rcu_sched
    9 ?        00:00:00 rcu_bh
   10 ?        00:00:00 migration/0
   11 ?        00:00:00 watchdog/0
   12 ?        00:00:00 cpuhp/0
   13 ?        00:00:00 cpuhp/1
   14 ?        00:00:00 watchdog/1
   15 ?        00:00:00 migration/1
   16 ?        00:00:00 ksoftirqd/1
   18 ?        00:00:00 kworker/1:0H
   19 ?        00:00:00 kdevtmpfs
   20 ?        00:00:00 netns
   21 ?        00:00:00 rcu_tasks_kthre
   22 ?        00:00:00 kauditd
   24 ?        00:00:00 khungtaskd
   25 ?        00:00:00 oom_reaper
16694 ?        00:00:00 xfsalloc
16695 ?        00:00:00 xfs_mru_cache
16707 ?        00:00:00 jfsIO
16708 ?        00:00:00 jfsCommit
16709 ?        00:00:00 jfsCommit
16710 ?        00:00:00 jfsSync
23503 ?        00:00:00 deja-dup
23569 ?        00:00:00 http
23570 ?        00:00:00 http
23617 ?        00:00:00 kworker/1:3
23765 ?        00:00:00 cupsd
23920 ?        00:00:00 gnome-terminal-
23927 pts/1    00:00:00 bash
24045 pts/1    00:00:00 mainprocess
24046 pts/1    00:00:00 mainprocess
24047 pts/1    00:00:00 mainprocess
24048 pts/1    00:00:00 subprocess1
24049 pts/1    00:00:00 subprocess1
24050 pts/1    00:00:00 subprocess2
24051 pts/1    00:00:00 subprocess2
24052 pts/1    00:00:00 subprocess2
24057 ?        00:00:00 sleep
24058 pts/1    00:00:00 ps
```

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -ef
```

Output

```
UID          PID  PPID  C STIME TTY          TIME CMD
root           1     0  0 13:37 ?        00:00:02 /sbin/init splash
root           2     0  0 13:37 ?        00:00:00 [kthreadd]
root           3     2  0 13:37 ?        00:00:00 [kworker/0:0]
root           4     2  0 13:37 ?        00:00:00 [kworker/0:0H]
root           6     2  0 13:37 ?        00:00:00 [mm_percpu_wq]
root           7     2  0 13:37 ?        00:00:00 [ksoftirqd/0]
root           8     2  0 13:37 ?        00:00:00 [rcu_sched]
root           9     2  0 13:37 ?        00:00:00 [rcu_bh]
root          10     2  0 13:37 ?        00:00:00 [migration/0]
root          11     2  0 13:37 ?        00:00:00 [watchdog/0]
root          12     2  0 13:37 ?        00:00:00 [cpuhp/0]
root          13     2  0 13:37 ?        00:00:00 [cpuhp/1]
root          14     2  0 13:37 ?        00:00:00 [watchdog/1]
root          15     2  0 13:37 ?        00:00:00 [migration/1]
root          16     2  0 13:37 ?        00:00:00 [ksoftirqd/1]
root          18     2  0 13:37 ?        00:00:00 [kworker/1:0H]
root          19     2  0 13:37 ?        00:00:00 [kdevtmpfs]
root          20     2  0 13:37 ?        00:00:00 [netns]
root          21     2  0 13:37 ?        00:00:00 [rcu_tasks_kthre]
root          22     2  0 13:37 ?        00:00:00 [kauditd]
root          24     2  0 13:37 ?        00:00:00 [khungtaskd]
root          25     2  0 13:37 ?        00:00:00 [oom_reaper]
root          26     2  0 13:37 ?        00:00:00 [writeback]
root       16694     2  0 14:12 ?        00:00:00 [xfsalloc]
root       16695     2  0 14:12 ?        00:00:00 [xfs_mru_cache]
root       16707     2  0 14:12 ?        00:00:00 [jfsIO]
root       16708     2  0 14:12 ?        00:00:00 [jfsCommit]
root       16709     2  0 14:12 ?        00:00:00 [jfsCommit]
root       16710     2  0 14:12 ?        00:00:00 [jfsSync]
onworks    23503   830  0 14:12 ?        00:00:00 deja-dup --prompt
_apt       23569  1984  0 14:12 ?        00:00:00 /usr/lib/apt/methods/http
_apt       23570  1984  0 14:12 ?        00:00:00 /usr/lib/apt/methods/http
root       23617     2  0 14:12 ?        00:00:00 [kworker/1:3]
root       23765     1  0 14:13 ?        00:00:00 /usr/sbin/cupsd -l
onworks    23920   830  0 14:13 ?        00:00:00 /usr/lib/gnome-terminal/gnome-te
onworks    23927 23920  0 14:13 pts/1    00:00:00 bash
onworks    24045 23927  0 14:17 pts/1    00:00:00 ./mainprocess
onworks    24046 24045  0 14:17 pts/1    00:00:00 ./mainprocess
onworks    24047 24045  0 14:17 pts/1    00:00:00 ./mainprocess
onworks    24048 24046  0 14:17 pts/1    00:00:00 ./subprocess1
onworks    24049 24046  0 14:17 pts/1    00:00:00 ./subprocess1
onworks    24050 24047  0 14:17 pts/1    00:00:00 ./subprocess2
onworks    24051 24047  0 14:17 pts/1    00:00:00 ./subprocess2
onworks    24052 24047  0 14:17 pts/1    00:00:00 ./subprocess2
root       24083  5062  0 14:18 ?        00:00:00 sleep 5
onworks    24084 23927  0 14:18 pts/1    00:00:00 ps -ef
```

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
`onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -ef | grep subprocess`
Output

```
onworks   1699  1633  0 13:37 ?        00:00:00 /usr/lib/evolution/evolution-cal
endar-factory-subprocess --factory contacts --bus-name org.gnome.evolution.datas
erver.Subprocess.Backend.Calendarx1633x2 --own-path /org/gnome/evolution/dataser
ver/Subprocess/Backend/Calendar/1633/2
onworks   1732  1633  0 13:37 ?        00:00:00 /usr/lib/evolution/evolution-cal
endar-factory-subprocess --factory local --bus-name org.gnome.evolution.dataserv
er.Subprocess.Backend.Calendarx1633x3 --own-path /org/gnome/evolution/dataserver
/Subprocess/Backend/Calendar/1633/3
onworks   1786  1729  0 13:37 ?        00:00:00 /usr/lib/evolution/evolution-add
ressbook-factory-subprocess --factory local --bus-name org.gnome.evolution.datas
erver.Subprocess.Backend.AddressBookx1729x2 --own-path /org/gnome/evolution/data
server/Subprocess/Backend/AddressBook/1729/2
onworks  24114 24113  0 13:48 pts/1    00:00:00 ./subprocess2
onworks  24115 24113  0 13:48 pts/1    00:00:00 ./subprocess2
onworks  24116 24113  0 13:48 pts/1    00:00:00 ./subprocess2
onworks  24117 24112  0 13:48 pts/1    00:00:00 ./subprocess1
onworks  24118 24112  0 13:48 pts/1    00:00:00 ./subprocess1
onworks  24427 23960  0 13:55 pts/1    00:00:00 grep --color=auto subprocess
```

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
`onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -eo pid,user,pcpu,pmem,cmd`
Output

```
  PID USER      %CPU %MEM CMD
    1 root       0.1  0.1 /sbin/init splash
    2 root       0.0  0.0 [kthreadd]
    3 root       0.0  0.0 [kworker/0:0]
    4 root       0.0  0.0 [kworker/0:0H]
    6 root       0.0  0.0 [mm_percpu_wq]
    7 root       0.0  0.0 [ksoftirqd/0]
    8 root       0.0  0.0 [rcu_sched]
    9 root       0.0  0.0 [rcu_bh]
   10 root       0.0  0.0 [migration/0]
   11 root       0.0  0.0 [watchdog/0]
   12 root       0.0  0.0 [cpuhp/0]
   13 root       0.0  0.0 [cpuhp/1]
   14 root       0.0  0.0 [watchdog/1]
   15 root       0.0  0.0 [migration/1]
   16 root       0.0  0.0 [ksoftirqd/1]
   18 root       0.0  0.0 [kworker/1:0H]
   19 root       0.0  0.0 [kdevtmpfs]
   20 root       0.0  0.0 [netns]
   21 root       0.0  0.0 [rcu_tasks_kthre]
   22 root       0.0  0.0 [kauditd]
   24 root       0.0  0.0 [khungtaskd]
   25 root       0.0  0.0 [oom_reaper]
   26 root       0.0  0.0 [writeback]
16694 root       0.0  0.0 [xfsalloc]
16695 root       0.0  0.0 [xfs_mru_cache]
16707 root       0.0  0.0 [jfsIO]
16708 root       0.0  0.0 [jfsCommit]
16709 root       0.0  0.0 [jfsCommit]
16710 root       0.0  0.0 [jfsSync]
23503 onworks    0.0  0.9 deja-dup --prompt
23569 _apt       0.0  0.1 /usr/lib/apt/methods/http
23570 _apt       0.0  0.1 /usr/lib/apt/methods/http
23617 root       0.0  0.0 [kworker/1:3]
23765 root       0.0  0.2 /usr/sbin/cupsd -l
23920 onworks    0.2  1.0 /usr/lib/gnome-terminal/gnome-terminal-server
23927 onworks    0.0  0.1 bash
24045 onworks    0.0  0.0 ./mainprocess
24046 onworks    0.0  0.0 ./mainprocess
24047 onworks    0.0  0.0 ./mainprocess
24048 onworks    0.0  0.0 ./subprocess1
24049 onworks    0.0  0.0 ./subprocess1
24050 onworks    0.0  0.0 ./subprocess2
24051 onworks    0.0  0.0 ./subprocess2
24052 onworks    0.0  0.0 ./subprocess2
24124 root       0.0  0.0 sleep 5
24125 onworks    0.0  0.1 ps -eo pid,user,pcpu,pmem,cmd
```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
`onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem`
Output

```
  PID USER      %CPU %MEM CMD
 1601 onworks    2.4  6.1 compiz
 1984 root       0.1  2.1 /usr/bin/python3 /usr/sbin/aptd
  812 root       1.0  2.0 /usr/lib/xorg/Xorg -core :0 -seat seat0 -auth /var/run/
 1660 onworks    0.1  2.0 /usr/bin/gnome-software --gapplication-service
 1054 onworks    0.0  1.5 /usr/lib/i386-linux-gnu/bamf/bamfdaemon
 1633 onworks    0.0  1.5 /usr/lib/evolution/evolution-calendar-factory
 1684 onworks    0.0  1.2 nautilus -n
 1699 onworks    0.0  1.2 /usr/lib/evolution/evolution-calendar-factory-subproces
 1732 onworks    0.0  1.1 /usr/lib/evolution/evolution-calendar-factory-subproces
 1449 onworks    0.0  1.0 /usr/lib/i386-linux-gnu/hud/hud-service
23920 onworks    0.2  1.0 /usr/lib/gnome-terminal/gnome-terminal-server
 1654 onworks    0.0  1.0 nm-applet
23503 onworks    0.0  0.9 deja-dup --prompt
 1451 onworks    0.0  0.9 /usr/lib/unity-settings-daemon/unity-settings-daemon
 1468 onworks    0.0  0.9 /usr/lib/i386-linux-gnu/unity/unity-panel-service
 1085 onworks    0.0  0.9 /usr/lib/ibus/ibus-ui-gtk3
 1520 onworks    0.0  0.8 /usr/lib/i386-linux-gnu/indicator-keyboard/indicator-ke
 1958 onworks    0.0  0.7 update-notifier
 1522 onworks    0.0  0.7 /usr/lib/i386-linux-gnu/indicator-printers/indicator-pr
 2055 onworks    0.0  0.7 /usr/bin/unity-scope-loader applications/applications.s
 1670 onworks    0.0  0.6 /usr/lib/policykit-1-gnome/polkit-gnome-authentication-
 1091 onworks    0.0  0.6 /usr/lib/ibus/ibus-x11 --kill-daemon
 1602 onworks    0.0  0.6 /usr/lib/evolution/evolution-source-registry
```

```
   42 root       0.0  0.0 [ecryptfs-kthrea]
   84 root       0.0  0.0 [kthrotld]
   85 root       0.0  0.0 [acpi_thermal_pm]
   87 root       0.0  0.0 [scsi_eh_0]
   88 root       0.0  0.0 [scsi_tmf_0]
   89 root       0.0  0.0 [scsi_eh_1]
   90 root       0.0  0.0 [scsi_tmf_1]
   91 root       0.0  0.0 [kworker/u4:2]
   93 root       0.0  0.0 [ipv6_addrconf]
  103 root       0.0  0.0 [kstrp]
  120 root       0.0  0.0 [charger_manager]
  171 root       0.1  0.0 [kworker/1:1H]
  172 root       0.0  0.0 [kworker/0:1H]
  196 root       0.1  0.0 [jbd2/sda1-8]
  197 root       0.0  0.0 [ext4-rsv-conver]
  239 root       0.0  0.0 [kworker/0:2]
16694 root       0.0  0.0 [xfsalloc]
16695 root       0.0  0.0 [xfs_mru_cache]
16707 root       0.0  0.0 [jfsIO]
16708 root       0.0  0.0 [jfsCommit]
16709 root       0.0  0.0 [jfsCommit]
16710 root       0.0  0.0 [jfsSync]
23617 root       0.0  0.0 [kworker/1:3]
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ ps -ejH | grep -E 'mainprocess
|subprocess'
```

Output

```
24045 24045 23927 pts/1    00:00:00           mainprocess
24046 24045 23927 pts/1    00:00:00             mainprocess
24048 24045 23927 pts/1    00:00:00               subprocess1
24049 24045 23927 pts/1    00:00:00               subprocess1
24047 24045 23927 pts/1    00:00:00             mainprocess
24050 24045 23927 pts/1    00:00:00               subprocess2
24051 24045 23927 pts/1    00:00:00               subprocess2
24052 24045 23927 pts/1    00:00:00               subprocess2
```

Question 7

Use `pstree` command with option that show the number of threads to each process.

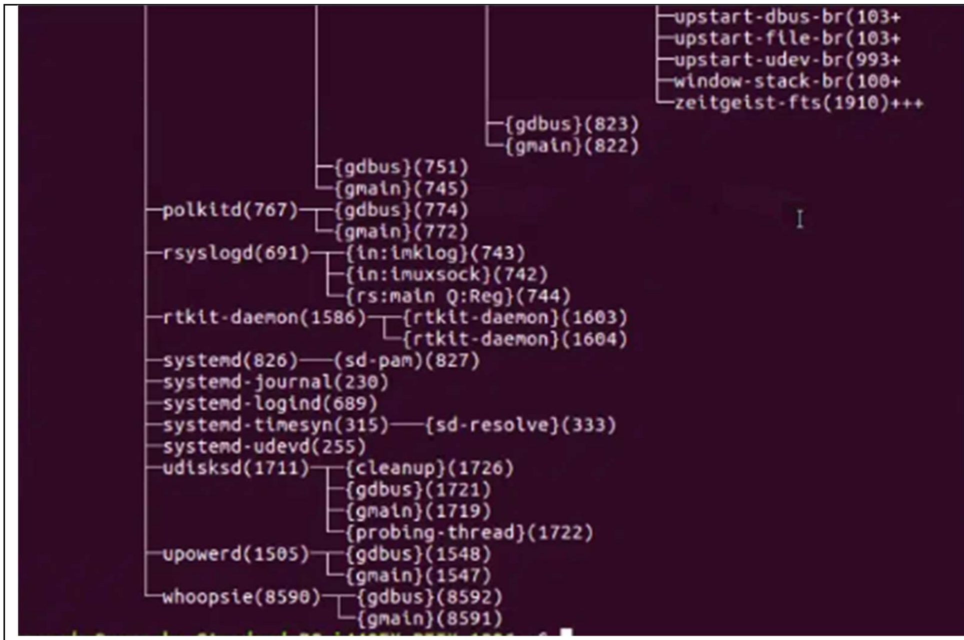
Command

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ pstree -p
```

Output

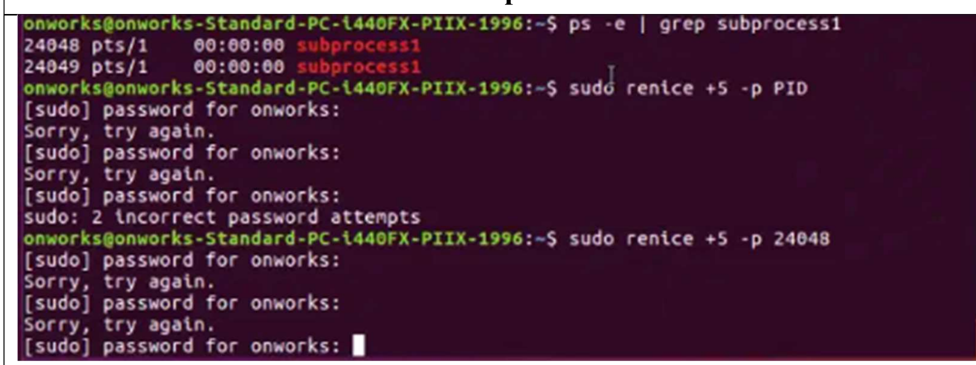
```
pstree: invalid option -- 'T'
Usage: pstree [ -a ] [ -c ] [ -h | -H PID ] [ -l ] [ -n ] [ -p ] [ -g ] [ -u ]
              [ -A | -G | -U ] [ PID | USER ]
       pstree -V
Display a tree of processes.

  -a, --arguments     show command line arguments
  -A, --ascii         use ASCII line drawing characters
  -c, --compact       don't compact identical subtrees
  -h, --highlight-all highlight current process and its ancestors
  -H PID,
  --highlight-pid=PID highlight this process and its ancestors
  -g, --show-pgids    show process group ids; implies -c
  -G, --vt100         use VT100 line drawing characters
  -l, --long          don't truncate long lines
  -n, --numeric-sort  sort output by PID
  -N type,
  --ns-sort=type      sort by namespace type (ipc, mnt, net, pid, user, uts)
  -p, --show-pids     show PIDs; implies -c
  -s, --show-parents  show parents of the selected process
  -S, --ns-changes    show namespace transitions
  -u, --uid-changes   show uid transitions
  -U, --unicode       use UTF-8 (Unicode) line drawing characters
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ pstree -p
systemd(1)─┬─NetworkManager(685)─┬─dhclient(803)
           │                     ├─dnsmasq(866)
           │                     ├─{gdbus}(759)
           │                     └─{gmain}(757)
           ├─accounts-daemon(657)─┬─{gdbus}(750)
           │                      └─{gmain}(738)
           ├─acpid(649)
           ├─agetty(1208)
           ├─aptd(1984)─┬─http(24326)
           │            ├─http(24327)
           │            └─{gmain}(1986)
           ├─avahi-daemon(653)───avahi-daemon(697)
           ├─colord(1652)─┬─{gdbus}(1658)
           │              └─{gmain}(1656)
           ├─cron(660)
           ├─cupsd(23765)
           ├─dbus-daemon(650)
           ├─irqbalance(747)
           ├─lightdm(741)─┬─Xorg(812)─┬─{InputThread}(818)
           │              │           ├─{llvmpipe-0}(814)
           │              │           └─{llvmpipe-1}(815)
           │              ├─lightdm(821)─┬─upstart(830)─┬─at-spi-bus-laun(108+
           │              │              │              ├─at-spi2-registr(110+
           │              │              │              ├─bamfdaemon(1054)─┬─+
           │              │              │              │                  ├─+
           │              │              │              │                  ├─+
           │              │              │              │                  ├─+
           │              │              │              │                  └─+
```



Question 8

Use `renice` command to change priority level of one of process "subprocess1".

Command
Output


Question 9

Terminate all running processes with the name “mainprocess”.

Command
`onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ pkill mainprocess`
Output

`Main process (ID: 24045) received signal: 15. Terminating...`
`Main process (ID: 24046) received signal: 15. Terminating...`
`Main process (ID: 24047) received signal: 15. Terminating...`
`[1]+  Done                    ./mainprocess`

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
import os

def main():
        pid = os.fork()

        if pid == 0:
                print(f"Child Process: PID = {os.getpid()}, Parent PID = {os.getpid()}")

        else:
                print(f"Parent Process: PID = {os.getpid()}, waiting for child...")
                os.wait()
                print("Parent: Child process has completed.")

        __name__ == "__main__":
        main()
```

Output:

```
onworks@onworks-Standard-PC-i440FX-PIIX-1996:~$ python3 fork_example.py
  File "fork_example.py", line 7
    print(f"Child Process: PID = {os.getpid()}, Parent PID = {os.getpid()}")
                                                                           ^
SyntaxError: invalid syntax
```